

Besonin Aqua Nasal Spray 64mcg/dose

budesonide

Nasal spray, suspension

Qualitative and quantitative composition

One dose (0.05 ml) contains budesonide 64 micrograms.

Product Description

Besonin Aqua Nasal Spray is a metered-dose, manual-pump spray formulation containing a micronized suspension of budesonide in an aqueous medium. It appears as a white to almost white suspension.

Therapeutic indications

Seasonal allergic and allergic/non-allergic perennial rhinitis. Treatment of nasal polyps. Prevention of nasal polyps after polypectomy.

Dosage and method of administration

The posology must be adjusted individually.

Rhinitis

Adults and children from 6 years: The recommended initial dose is 256 micrograms daily. The dose can be administered once daily in the morning or divided into two administrations, morning and evening. This means 128 micrograms (2 X 64 micrograms) into each nostril in the morning or 64 micrograms into each nostril morning and evening.

No further effect has been seen with doses higher than 256 micrograms per day.

For elderly patients the posology is the same as for adults.

When the desired clinical effect has been achieved, the dose should be reduced to the lowest amount that is needed to control the symptoms. A dose of 32 micrograms in each nostril in the morning may be sufficient for some patients.

In some patients relief of symptoms is already obtained within 5 – 7 hours after the start of treatment. The full effect is only obtained after several days of treatment (in rare cases not until after 2 weeks). Treatment of seasonal rhinitis should therefore start before exposure to the allergens.

In case of severe nasal congestion an initial addition of a vasoconstrictor may be required.

Supplementary treatment may sometimes be necessary in order to counteract possible ocular symptoms caused by the allergy.

Symptomatic treatment and prevention of nasal polyps

The recommended dose is 256 micrograms daily. The dose may be administered once daily in the morning or divided into two administrations morning and evening. When the desired clinical effect

has been achieved, the dose should be reduced to the lowest amount that is needed to control the symptoms.

Instructions for correct use of Besonin Aqua:

It is important to instruct the patient to carefully read: "Instruction for use and handling , and disposal"

Contraindications

Previous hypersensitivity to budesonide or to any of the excipients.

Special warnings and special precautions for use

In long-term treatment with high doses, systemic effects of glucocorticosteroids such as hypercortisolism, adrenal suppression and/or delayed growth in children can occur. Long-term effects of nasal steroids in children have not been elucidated. Treatment with medicines containing cortisone can lead to slower growth. Regular monitoring of growth is recommended in children and adolescents receiving long-term treatment with corticosteroids, irrespective of the administration form. If delayed growth is suspected, this should be investigated. The benefits of glucocorticosteroid treatment should be placed in relation to possible risks of inhibition of growth.

Caution should be observed when treating patients with nasal fungal or herpes infections.

Caution must be exercised in the treatment of patients who are being transferred from systemically acting glucocorticosteroids to Besonin Aqua and if there is suspected disturbed pituitary-adrenocortical function. In these patients there should be a careful reduction of the systemic steroid dose and testing of hypothalamic-pituitary-adrenocortical function should be considered. They may also require the addition of systemic steroids in connection with periods of stress, e.g. surgery, trauma, etc.

Severely impaired hepatic function affects the pharmacokinetics of orally administered budesonide, resulting in a reduced rate of elimination and increased systemic availability. Consideration may need to be given to possible systemic effects.

Special care may be needed in patients with lung tuberculosis.

Besonin Aqua should not come into contact with the eyes. If Besonin Aqua comes into contact with the eyes, rinse immediately with water.

Concomitant treatment with ketoconazole or other potent CYP3A4-inhibiting drugs should be avoided. If this is not possible, the interval between administrations of the agents should be as long as possible (see Interactions with other medicinal products and other forms of interaction).

Interactions with other medicinal products and other forms of interaction

Oral ketoconazole 200 mg once daily increased the plasma concentrations of oral budesonide (3 mg in a single dose) on average six-fold when administered simultaneously. When oral ketoconazole was administered 12 hours after budesonide, the concentration increased on average three-fold. There is no information about this interaction for nasal budesonide, but also in such cases greatly increased plasma levels are expected. Since there are no data to enable dosage recommendations to be given in cases of nasal administration the combination should be avoided. If this is not possible,

the time interval between administration of ketoconazole and that of budesonide should be as long as possible. A reduction of the budesonide dose must also be considered. Other potent inhibitors of CYP3A4 also probably cause a marked increase in the plasma levels of budesonide.

Pregnancy and lactation

Pregnancy

There is no increased risk of congenital defects in infants born to mothers treated with Besonin Aqua. A slightly increased occurrence of slight heart malformation compared to what might be expected has been confirmed in infants whose mothers were exposed to Besonin Aqua in early pregnancy, but a causal connection with the exposure is considered improbable.

During pregnancy the aim should be the lowest effective dose and the shortest possible treatment time with Besonin Aqua.

Lactation

Budesonide is excreted in breast milk. However, at therapeutic doses of Besonin Aqua no effects on the suckling child are anticipated. Besonin Aqua can be used during breast-feeding.

Effects on ability to drive and use machines

Besonin Aqua does not affect ability to drive or operate machinery.

Undesirable effects

Approx. 5% of treated patients may be expected to experience side effects in the form of local irritation.

In below table the undesirable effects are listed according to classification and frequency.

Common, Uncommon, Rare, Very rare

<i>Respiratory, thoracic and mediastinal disorders</i>	Common	Local irritation, epistaxis, slight haemorrhagic nasal secretion
	Very rare	Nasal septum perforation, ulceration of mucous membranes
<i>Immune system disorders</i>	Uncommon	Angioedema, urticaria, dermatitis, rash, pruritus
	Very rare	Anaphylactic reaction

Immediate or delayed hypersensitivity reactions including urticaria, rash, dermatitis angioedema and pruritus have been reported.

In very rare cases, ulceration of mucous membrane and nasal septum perforation have occurred during use of nasally applied steroids. The cause of these side effects (the steroid, the underlying disease, or other factors) is not clear.

Overdose

Acute overdosage with Besonin Aqua, even high doses, is not expected to cause any clinical problems. If Besonin Aqua is used in high doses for a long period, the systemic effects characteristic of glucocorticosteroids such as hypercortisolism and adrenal suppression may occur.

Pharmacodynamic properties

Pharmacotherapeutic group: Glucocorticoids

ATC code: R01AD05

Budesonide is a glucocorticosteroid with powerful local anti-inflammatory effects. The precise mechanism of action of glucocorticosteroid in the treatment of rhinitis is not fully understood. Anti-inflammatory effects such as inhibited release of inflammatory mediators and inhibition of cytokine-mediated immune response are probably significant. The activity of budesonide measured as the affinity for glucocorticosteroid receptors is approx. 15 times higher than that of prednisolone.

Budesonide given prophylactically prior to nasal provocation has proved to protect against immigration of eosinophils and hyperreactivity.

At recommended doses, Besonin Aqua does not cause any clinically significant changes either in basal plasma cortisol levels or in the response to stimulation with ACTH. However, dose-related suppression of plasma and urinary cortisol has been observed following short-term administration of Besonin Aqua.

A dose-response relationship has not been demonstrated in children with seasonal or perennial allergic rhinitis or in adults with perennial allergic rhinitis.

Pharmacokinetic properties

Absorption

The systemic availability of budesonide from Besonin Aqua is 33% of the measured dose. The kinetics are proportional to the dose in the therapeutic dose range.

The peak plasma concentration in adults from 256 micrograms of budesonide from Besonin Aqua is 0.64 nmol/L, and is reached within 0.7 hours. The AUC (area under the curve) after administration of 256 micrograms of budesonide from Besonin Aqua is 2.7 nmol x hours/litre in adults and 5.5 nmol x hours/litre in children, which indicates a higher systemic exposure to glucocorticoids in children.

Distribution and metabolism

Budesonide has a volume of distribution of approx.. 3 l/kg. Binding to plasma proteins is approx.. 85-90 %.

Budesonide undergoes extensive (~90%) first pass metabolism in the liver to metabolites with low glucocorticosteroid activity. The glucocorticosteroid activity of the principal metabolites, 6-beta-hydroxybudesonide and 16-alpha-hydroxyprednisolone, is less than 1% of that of budesonide. Budesonide does not undergo local metabolism in the nose.

Elimination

Budesonide is eliminated by metabolism which is principally catalysed by the enzyme CYP3A4. The metabolites are excreted in the urine in unchanged or conjugated form. Only negligible amounts are found unchanged in the urine. Budesonide has high systemic clearance (~1.2 l/min) and the half-life in plasma after intravenous administration is on average approx. 4 hours.

List of excipients

Potassium sorbate
Polysorbate 80
Dextrose Anhydrous
Edetate disodium
Hydrochloric acid
Microcrystalline Cellulose and Carboxymethyl Cellulose Sodium
Purified water

Special precautions for storage

Do not store above 30°C.
Do not freeze.

Pack size

120 doses

Manufactured by

SYNMOSA BIOPHARMA CORPORATION, Synmosa Plant
No.6, Kuang Yeh 1st Road, Hu-Kuo Hsiang, Hsin-Chu Ind. Park, Hsin Chu Hsien, Taiwan, R.O.C.

Product Registration Holder

RIGEL PHARMA SDN. BHD.
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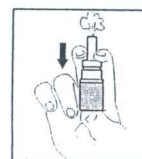
Date of Revision

19 May 2016

Instructions for use and handling, and disposal

Before Besonin Aqua is used for the first time, the nasal applicator must be loaded with the drug. Shake the bottle, and then spray into the air, until there is an even spray. The effect of this lasts approx. 24 hours. If a longer time elapses before the next dose is taken, the nasal applicator must be loaded with the drug again. This time it is sufficient to spray just once (1) into the air.

1. Blow your nose.
Shake the bottle.
Remove the protective cap.
2. Hold the bottle as shown in the pictures.
3. Insert the tip into your nostril and pump the number of doses prescribed. Administer the spray into the other nostril in the same way.
4. Replace the protective cap. Do not use Besonin Aqua nasal spray more often than prescribed.

**Children**

Children should only use Besonin Aqua nasal spray under supervision by an adult to ensure that the dose is correctly administered and that the dosage is according to the doctor's prescription.

Cleaning

Clean the upper plastic parts regularly. Remove the protective cap and lift off the white nasal applicator. Wash the plastic parts in warm water. Allow the plastic part to air-dry completely before replacing them.